

D SERIES

SINGLE-PHASE SOLID-STATE RELAY

Built in TVS protection, IP20 protection level, standard size 22.5mm rail installation

- Application scenarios
- Mainly used in glass, photovoltaics, etc



Product Introduction

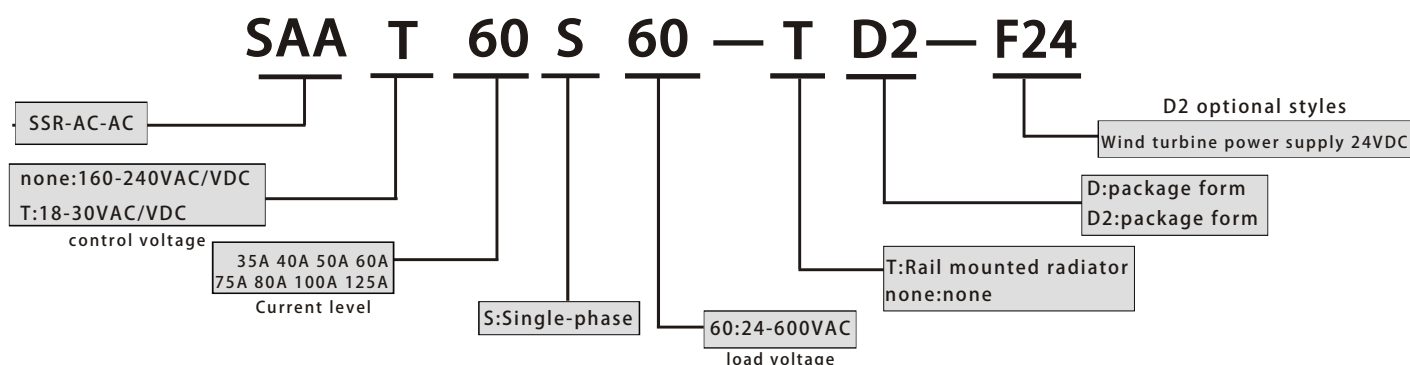
The D package series is a high-power rail mounted AC solid-state relay designed specifically for industrial control applications, with built-in TVS protection. The current ranges from 35-120A, the load voltage range is 24-600VAC, and the control voltage is 160-240VAC/VDC and 18-30VAC/VDC.

Using ultra-thin heat sink integrated solid-state relays, this ultra-narrow solid-state relay represents an evolution in **SENNDIK**'s expertise in solid-state switching. Our ultra-thin heat sink integrated solid-state relay has a width half that of the K-type solid-state relay, saving 50% of installation space, reducing installation time, and enabling faster and easier wiring without any impact on the performance or reliability of the solid-state relay. The D-packaged series solid-state relays, due to their compact size, offer a unique opportunity to conserve panel space.

Has the following characteristics:

- TVS protection
- IP20 protection level
- Photoelectric isolation
- EMC anti-interference design
- Integrated heat dissipation and packaging
- You can choose communication control
- Fourth generation TMS square process
- Can install 35mm standard guide rail (DINBN50022) can also fix screws

SELECTION GUIDE

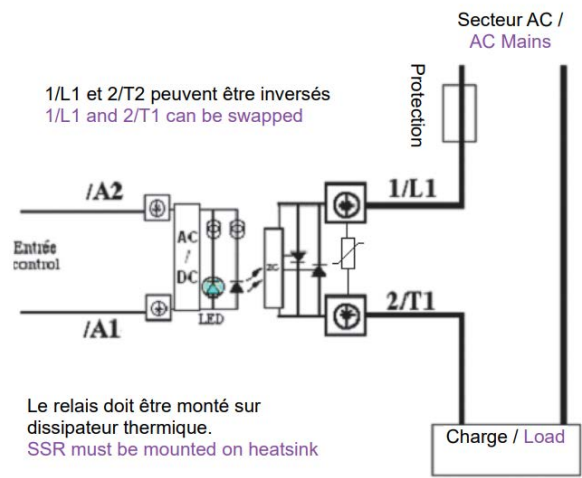


Product parameters (input TA=25 °)		
Control voltage range	160-240VAC/VDC	18-30VAC/VDC
Connected voltage	160VAC/VDC	18VAC/VDC
Turn off voltage	5VAC/VDC	3VAC/VDC
Maximum input current	6.7ma (@240VAC/50HZ)	9.2ma(@30VAC/50HZ)

Output parameters (TA=25 ° C)								
Working voltage range		S60:24-600VAC						
Off state leakage current		1ma						
Minimum load current		0.6ma						
Start time	Zero crossing type	1/2 cycle+1ms						
	Random type	1ms						
Closing Time	Zero crossing type	1/2 cycle+1ms						
	Random type	1ms						
Maximum I ² t (10ms. A ² s)		3200	3200	4050	4050	7800	11250	11250
Maximum output voltage drop		1.7V r.m.s						
Minimum critical rise rate of off state voltage dv/dt		1000[V/us]min						
Transient voltage VPK		1600[VPK]						
Input and output terminals, insulation voltage between output and substrate (50/60Hz)					>2000Vrms			
Working environment temperature range		-40℃ ~+80℃						
Storage environment temperature range		-40℃ ~+100℃						

Other parameters	
frequency range	47H-63Hz
insulation resistance	1000Mo Ω (500VDC)
dielectric withstand	4000Vrms,1 minute,Between input and output
	2500Vrms,1 minute,Between input, output, and radiator
Working environment temperature range	-30 °C ~+80 °C
Storage environment temperature range	-30 °C ~+100 °C

SCHEMATIC DIAGRAM OF WIRING



Precautions

- 1.Relays are power devices, and space should be left between relays during installation, as shown in Figure 1
- 2.The installation environment of the relay requires ventilation and heat dissipation, ensuring that the temperature of the relay radiator does not exceed 90 °C under any circumstances
- 3.It is necessary to use a screwdriver and wires as required to connect the relay, as shown in Figure 1
- 4.Do not use this product beyond the scope of the instruction manual

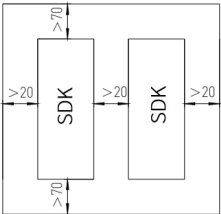


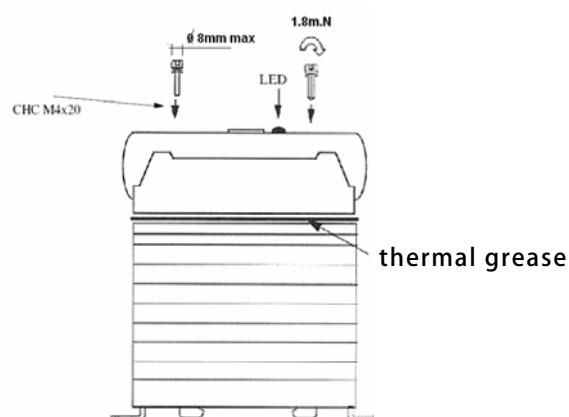
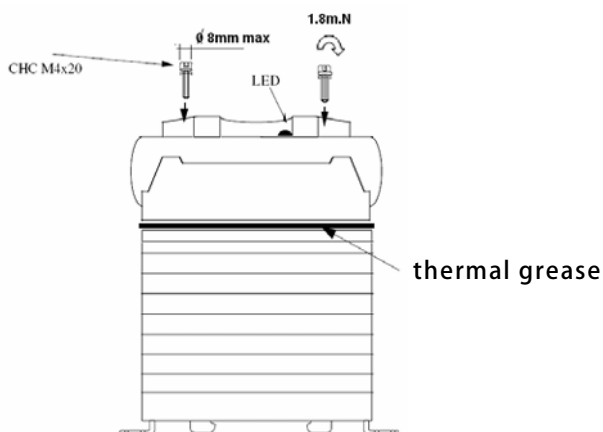
Figure 1 Installation distance requirements

		Input terminals A1, A2	Load end 1/L1, 2/T1
Screwdriver specifications		slotted screw,0.6x3.5mm	slotted screw2#,0.8x5.5mm
Recommended torque		0.6N.m	1N.m
Stripping length		7cm	10cm
Select wire diameter	single-core wire	1x0.5~2.5mm ² 2x0.5~1mm ²	1x0.5~2.5mm ² 2x0.5~1mm ²
	Multi core wire (with cold pressed terminals)	1x0.5~2.5mm ² 2x0.5~1mm ²	1x1.5~10mm ² 2x1.5~6mm ²
	American Standard Wire (AWG)	1x20 ~12	2x14 ~10

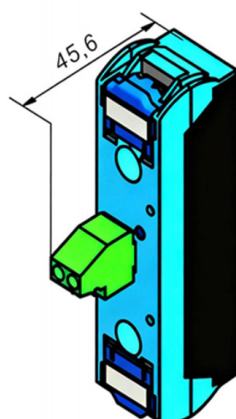
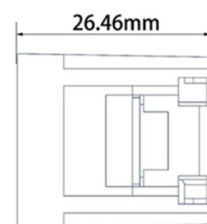
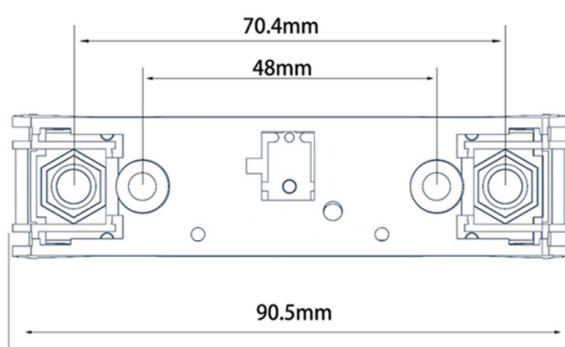
INSTALLATION INSTRUCTIONS

Use two screws, first tighten the two screws with a maximum torque of 1 Nm, and then continue to use the nominal torque of 1.8 Nm. It is recommended to use an electric screwdriver instead of a pneumatic tool. It can better adjust the specified tightening parameters, especially to achieve a smoother final torque. When using pneumatic systems, impact and torque overshoot may occur. Suggest limiting the speed of installing screws to ensure even flow and diffusion of thermal paste, especially when using denser thermal paste. If tightened at high speed, internal damage may occur. The maximum screw speed for tightening should not exceed 250 rpm. Soft tightening (without torque overshoot) will reduce internal stress.

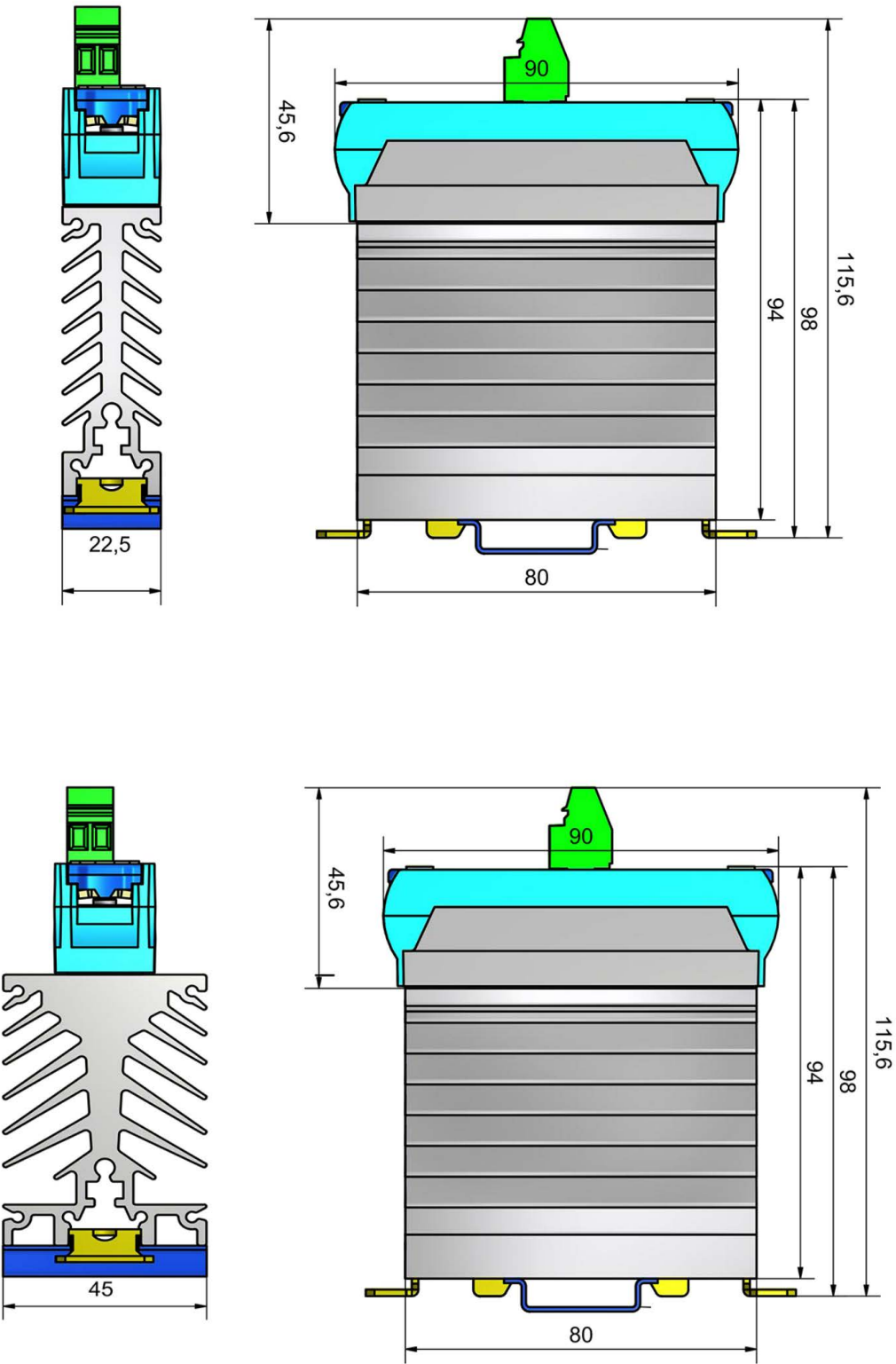
install



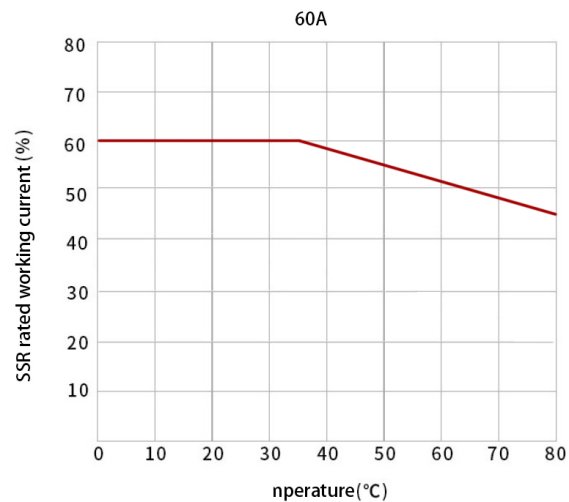
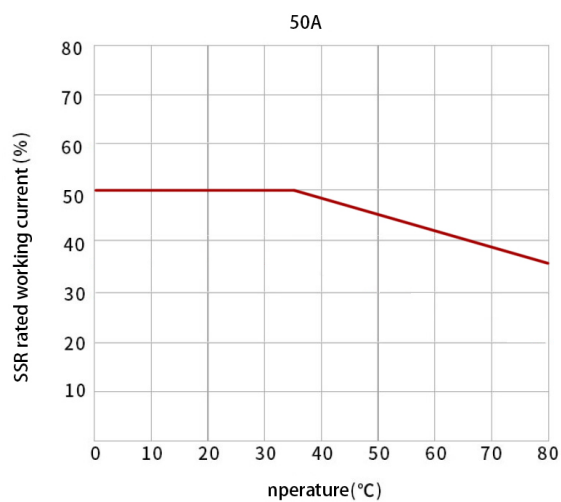
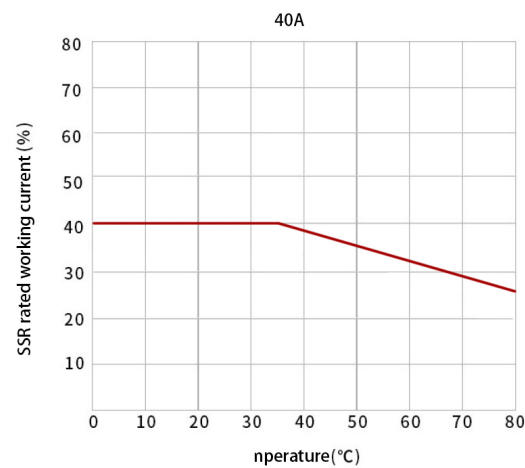
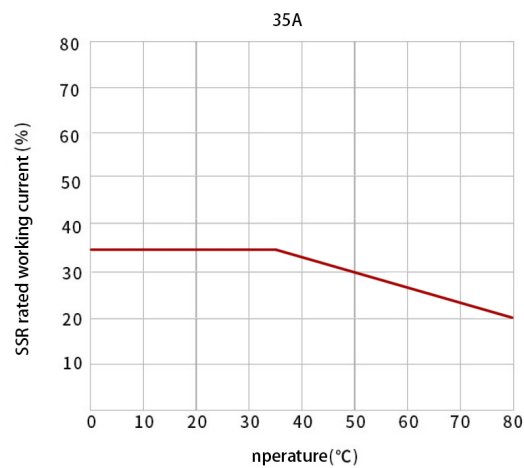
SIZE CHART



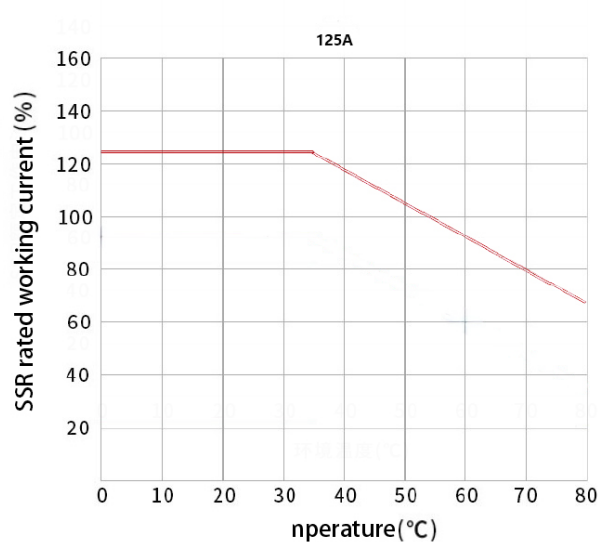
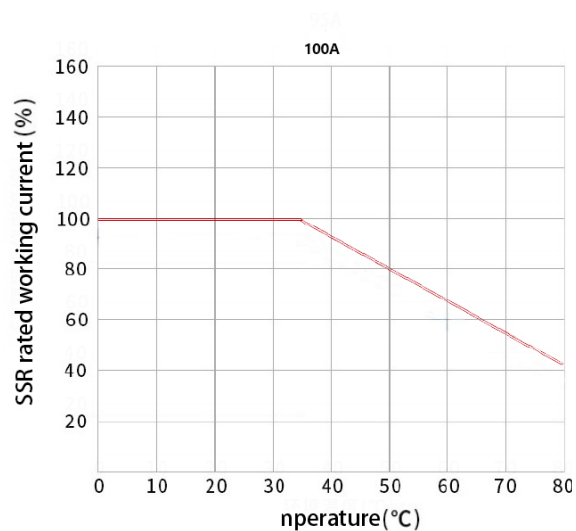
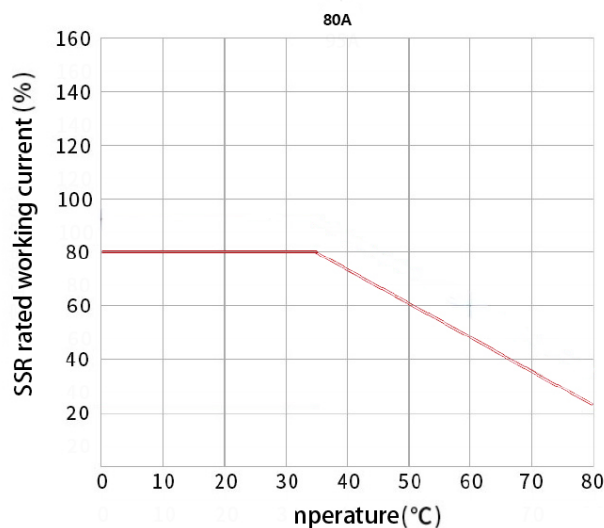
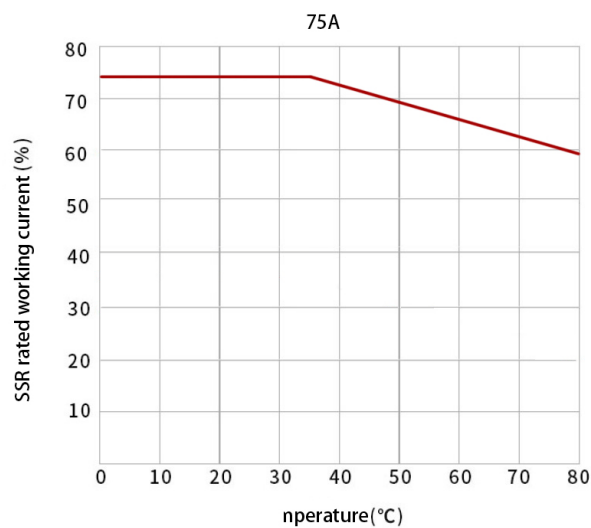
SIZE CHART



TEMPERATURE



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Any changes will not be notified separately.

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